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STATOR AND MOTOR UNIT INCLUDING SAME

Abstract

The stator includes a stator core having a circularly cylindrical shape, an oil guide (guide member) having a ring shape and arranged at least on one side in an axial direction of the stator core coaxially with the stator core, and coils attached to the stator core and the oil guide such that a plurality of coil ends of the coils axially protrude from the oil guide. The oil guide includes: an introduction part to which a cooling medium which is externally supplied is introduced; and a plurality of ejection parts which communicate with the introduction part and which are configured to eject the cooling medium toward the coil ends.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is based on JP application No. 2024-33419, filed Mar. 5, 2024, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a stator of a motor and a motor unit including the stator.

BACKGROUND OF THE DISCLOSURE

[0003] Along with recent popularization of electric vehicles, downsizing of a motor is required in terms of mounting of the motor, an increase in types, manufacturing, cost reduction, and the like of the electric vehicles. To downsize the motor without reducing power of the motor, a power density of the motor has to be increased, and more specifically, a current density of a current caused to flow thorough a coil has to be increased. However, when the current density is increased, the amount of heat generation of the coil increases, and therefore, for example, a technique for cooling the coil as disclosed in U.S. Ser. No. 11/462,958 becomes important.

[0004] In the technique disclosed in U.S. Ser. No. 11/462,958, however, an externally supplied cooling medium passes through an interior of the stator core and is then ejected toward the coil end. Therefore, the cooling medium which has drawn heat from the stator core and has increased in temperature is ejected onto the coil end. Such a technique disclosed in U.S. Ser. No. 11/462,958 is still susceptible to an improvement in efficiently cooling the coil.

SUMMARY

[0005] In view of the foregoing, it is an object of the present disclosure to provide: a stator configured such that a coil of the stator is efficiently cooled; and motor unit including the stator.

[0006] To achieve the object, a stator according to the present disclosure has the following configuration. The stator according to the present disclosure includes: a stator core having a circularly cylindrical shape; a guide member having a ring shape and arranged coaxially with the stator core on at least one side in an axial direction of the stator core; and coils attached to the stator core and the guide member such that a plurality of coil ends of the coils axially protrude from the guide member. The guide member includes: an introduction part to which a cooling medium which is externally supplied is introduced; and a plurality of ejection parts which communicate with the introduction part and which are configured to eject the cooling medium toward the plurality of coil ends.

[0007] With this configuration, the coils are attached to the stator core and the guide member such that the plurality of coil ends axially protrude from the guide member, and thereby, the guide member is arranged in the proximity of the coil ends of the coils. Further, since the guide member arranged as described above includes: the introduction part to which the externally supplied cooling medium is introduced; and the plurality of ejection parts which communicate with the introduction part and which are configured to eject the cooling medium toward the coil ends, the externally supplied cooling medium introduced into the introduction part is then ejected as it is toward the coil ends arranged in the proximity. Therefore, the externally supplied cooling medium is applied to the coil ends in a fresh state, a state where the externally supplied cooling medium has not drawn heat from another component of the stator. This provides the effect that the coils can be efficiently cooled.

[0008] Further, an accompanying effect is that configuring the guide member as a component separate from the stator core enables the guide member to be provided for the stator without influencing the strength and/or the manufacturability of the stator core.

[0009] In the stator, some of the plurality of coil ends may have tilted parts each extending obliquely outward in a direction intersecting the axial direction, and some of the plurality of ejection parts may be configured to eject the cooling medium toward inside parts of the tilted parts, the inside parts facing the guide member. With this configuration, the cooling medium is ejected toward the inside parts facing the guide member, that is, parts in the proximity of the guide

member, of the tilted parts of the coil ends. Thus, the cooling medium can reliably be applied to the coil ends.

[0010] In the stator, some of the plurality of ejection parts may be configured to eject the cooling medium radially inward from radially outside the plurality of coil ends. With this configuration, the cooling medium is ejected radially inward toward the coil ends located radially inside the ejection parts. In particular, toward the coil ends arranged on a side corresponding to the 6 o'clock side (a lower side) of the guide member, that is, the coil ends located upward of the ejection parts, the cooling medium is ejected upward against the gravity. Thus, the cooling medium can reliably be applied to the coil ends located radially inside the ejection parts.

[0011] In the stator, the stator core may have a plurality of first slots each of which axially penetrates the stator core and which are aligned side by side circumferentially, the guide member may have a plurality of second slots each of which axially penetrates the guide member such that the plurality of second slots correspond to the plurality of first slots, the plurality of coil ends may axially protrude from the plurality of second slots when the coils are inserted into the plurality of first slots and the plurality of second slots, and some of the plurality of ejection parts may be arranged between adjacent ones of the plurality of second slots. With this configuration, the ejection parts can be arranged in the proximity of the coil ends, and the cooling medium can be ejected from the proximity of the coil ends, regardless of the radial position of the coil ends of the coils inserted in the plurality of first slots and the plurality of second slots. Thus, the cooling medium can reliably be applied to the coil ends.

[0012] In the stator, the plurality of ejection parts may be configured to eject the cooling medium toward respective ones of the coil ends. Thus, the cooling medium can be applied to each of the coil ends, and thus, the entirety of the coils can be more efficiently cooled.

[0013] In the stator, the stator core may have an in-core flow path axially penetrating the stator core, and the guide member may have a core connector which communicates with the introduction part and which is connected to an inflow port of the in-core flow path. With this configuration, part of the externally supplied cooling medium is introduced into the introduction part and then flows from the core connector into the in-core flow path. Thus, the stator core can be cooled.

[0014] In the stator, the stator core may include a plurality of the guide members arranged on both sides in the axial direction of the stator core, both the guide members may communicate with each other via the in-core flow path such that the cooling medium passes through the in-core flow path from one of the guide members to the other of the guide members, the other guide member may be provided with a plurality of sub-ejection parts configured to eject the cooling medium flowing from the one guide member toward the coil ends in the proximity of the other guide member, that is, the coil ends on the other side. With this configuration, the cooling medium, which has passed through the in-core flow path in the stator core from the one guide member, is ejected from the sub-ejection parts of the other guide member toward the coil end on the other side. Thus, the cooling medium supplied for cooling the stator core can be used without waste to cool the coil ends on the other side.

[0015] In the stator, the stator core may have a plurality of first slots each penetrating the stator core in the axial direction, the other guide member may have a plurality of second slots corresponding, one by one, to the plurality of first slots, the coils may be inserted into the plurality of first slots and the plurality of second slots, each of the plurality of sub-ejection parts may be arranged radially outside the second slots of the other guide member, flow paths connected to the plurality of sub-ejection parts may have openings which are open toward the plurality of second slots. With this configuration, the coils inserted through the second slots face the flow paths via the openings, thereby increasing contact area between the cooling medium and the coils. Thus, the coils can be more efficiently cooled.

[0016] In the stator, the guide member may have a chamber including an annular part which is concentric, and the plurality of ejection parts may communicate with the introduction part via the

chamber. With this configuration, the cooling medium in the chamber can be ejected from the ejection parts in the state where the annular part is uniformly filled in the circumferential direction with the cooling medium introduced to the introduction part. This enables the cooling medium to be circumferentially uniformly ejected from the ejection parts, and thus, the cooling medium can be circumferentially uniformly applied to each of the coil ends.

[0017] In the stator, the guide member may be provided with a peripheral wall continuously surrounding the annular part from radially outside. With this configuration, a space radially outside the annular part is occupied by the peripheral wall, thereby reducing the volume of the flow path in the guide member. This can suppress air from accumulating in the guide member, thereby suppressing the ejection pressure of the cooling medium from the guide member from decreasing.

[0018] In the stator, the chamber may include a plurality of branches branched radially inward from the annular part, and the branches may have ends connected to the plurality of ejection parts. When the chamber includes the branches, this configuration eliminates a space in each branch between an end side and its part connected to a corresponding one of the ejection parts. This can suppress air from accumulating at the ends of the branches, thereby suppressing the ejection pressure of the cooling medium from the guide member from decreasing.

[0019] In the stator, the guide member may be made of non-magnetic material. This enables the guide member to be provided for the stator without magnetically influencing the motor.

[0020] In the stator, the guide member may be made of resin material. Thus, the degree of freedom of processing the guide member can be increased, and thus, the ejection part can be freely arranged depending on the arrangement of the coil ends.

[0021] The stator may include a sub-guide member arranged axially outside the guide member, being coaxial with the guide member, and having a cylindrical shape, the introduction part of the guide member may be open radially outward, and the sub-guide member may have an outer circumference part having a guide groove configured to receive the externally supplied cooling medium and guide the cooling medium to the introduction part. With this configuration, the externally supplied cooling medium is guided along the guide groove in the outer circumference part of the sub-guide member to the introduction part of the guide member. Thus, without increasing the size of the stator radially outward beyond the guide member, the externally supplied cooling medium can be guided to the introduction part, and thus, the cooling medium can be introduced to the introduction part with a simple and space-saving structure.

[0022] Moreover, a motor unit according to the present disclosure is a motor unit including: a motor including the stator; and a housing in which the stator is housed, and the housing is configured to press the guide member of the stator axially against the stator core. With this configuration, the cooling medium is suppressed from leaking from a location between the stator core and the guide member without providing a sealing member, such as an O-ring, between the stator core and the guide member. Thus, the number of components included in the motor unit can be reduced, and thus, the cost of the motor unit can be reduced.

[0023] As described above, the present disclosure enables the coils to be efficiently cooled.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] FIG. 1 is a schematic longitudinal sectional view of part of a motor unit of a first embodiment;

[0025] FIG. 2 is a perspective view of a stator of the first embodiment;

[0026] FIG. 3 is an exploded perspective view of the stator of FIG. 2;

[0027] FIG. 4 is an enlarged view of coil ends;

[0028] FIG. 5 is a schematic front view of part of an oil guide of the first embodiment;

[0029] FIG. **6** is a schematic perspective view of part of the oil guide of FIG. **5** viewed from an axially inner side;

[0030] FIG. **7** is a view illustrating a first flow pattern;

[0031] FIG. **8** is a view illustrating a flow of oil in the oil guide in the first flow pattern;

[0032] FIG. **9** is a view illustrating a second flow pattern;

[0033] FIG. **10** is a view illustrating a flow of oil in a front oil guide in the second flow pattern;

[0034] FIG. **11** is a view illustrating a flow of the oil in a rear oil guide in the second flow pattern;

[0035] FIG. **12** is a view illustrating a third flow pattern;

[0036] FIG. **13** is a view illustrating a flow of oil in the front oil guide in the third flow pattern;

[0037] FIG. **14** is a view illustrating a flow of the oil in the rear oil guide in the third flow pattern;

[0038] FIG. **15** is a view for explaining an ejection manner of ejection parts and sub-ejection parts;

[0039] FIG. **16** is an enlarged view of ejection parts on a radially outside and coil ends which are ejection targets of the ejection parts on the radially outside;

[0040] FIG. **17** is an enlarged view of ejection parts on a radially inside and coil ends which are ejection targets of the ejection parts on the radially inside;

[0041] FIG. **18** is an enlarged view of the sub-ejection parts and coil ends which are ejection targets of the sub-ejection parts;

[0042] FIG. **19** is a schematic longitudinal sectional view of a coil end arranged on a lower side of the oil guide and a peripheral part of the coil end;

[0043] FIG. **20** is a schematic front view of an oil guide of a second embodiment;

[0044] FIG. **21** is an enlarged view of an upper part of the oil guide of FIG. **20**; and

[0045] FIG. **22** is a schematic longitudinal sectional view of a flow path leading to a sub-ejection part of the second embodiment.

DETAILED DESCRIPTION

[0046] Embodiments of the present disclosure will be described with reference to the drawings below.

First Embodiment

Configuration of Motor Unit

[0047] FIG. **1** is a schematic longitudinal sectional view of part of a motor unit A of a first embodiment. Note that FIG. **1** shows one end in an axial direction of the motor unit A, and the other end in the axial direction appears in a similar manner. First of all, the configuration of the motor unit A according to the first embodiment of the present disclosure will be described. As shown in FIG. **1**, the motor unit A includes a motor **1** and a housing **9**.

[0048] The motor **1** is, for example, an interior permanent magnet motor (IPM motor) and is used as a driving source of an electric vehicle. The motor **1** includes: a shaft **10** which constitutes a rotational axis; a rotor **11** having a circularly cylindrical shape with a center hole in which the shaft **10** is fixed; and a stator **12** having a circularly cylindrical shape and arranged radially spaced apart from the rotor **11** such that an inner circumferential surface of the stator **12** faces an outer circumferential surface of the rotor **11**. The shaft **10** is rotatable together with the rotor **11** and relative to the stator **12**. The shaft **10**, the rotor **11**, and the stator **12** are arranged concentrically with one another, and axial directions thereof are collectively denoted by **6** in the figures. The configuration of the stator **12** will be described in detail later.

[0049] The motor **1** is supplied with oil as a cooling medium. More specifically, oil stored in an oil pan (not shown) and cooled by a heat exchanger (not shown) is pumped by an oil pump OP to the motor **1**. In FIG. **1**, a bold dashed arrow conceptually shows a flow of the oil, and likewise, bold arrows and bold dashed arrows in the following figures show flows of the oil. After used to cool the motor **1**, the oil is stored in the oil pan again.

[0050] The housing **9** houses the rotor **11** and the stator **12** while exposing both ends of the shaft **10**. The housing **9** has a flow path **90** formed therein. Through the flow path **90**, externally supplied oil flows to the motor **1**. The housing **9** may have another flow path (omitted in the figure) for the

oil, for example, a flow path through which the oil flows to another member (e.g., a bearing). The configuration of the housing **9** will be described in detail later.

Configuration of Stator

[0051] FIG. **2** is a perspective view of the stator **12** of the first embodiment. Subsequently, the configuration of the stator **12** according to the first embodiment of the present disclosure will be described in detail. As shown in FIG. **2**, the stator **12** includes a stator core **2**, an oil guide **3A** (a guide member), coils **4** each having a plurality of coil ends **40** at axial ends thereof, and a sub-oil guide **5** (sub-guide member). The stator **12** is configured to allow a current to flow from a power supply (not shown) via an input terminal **B** to the coils **4**, and the coils **4** serve as a main heat generation source of the stator **12**. Thus, to cool mainly the coils **4**, the stator **12** is configured to eject the oil flowing from the flow path **90** of the housing **9** toward the coil ends **40** of the coils **4** by using the oil guide **3A**. Each of the components of the stator **12** is first described below, and then, the detailed configuration of the oil guide **3A** will be described together with flow patterns of the oil.

Each Component of Stator

[0052] FIG. **3** is an exploded perspective view of the stator **12** of FIG. **2**. Note that in FIG. **3**, the coils **4** are not shown for the sake of illustration. As shown in FIG. **3**, the stator core **2** has a circularly cylindrical shape and is made of magnetic material such as magnetic steel. The stator core **2** includes a plurality of first teeth **21** formed over the entire axial length of the stator core **2**. The plurality of first teeth **21** extend circumferentially inward and are aligned side by side circumferentially. A space between adjacent ones of the plurality of first teeth **21** is a first slot **20**. Thus, the stator core **2** has a plurality of first slots **20** axially penetrating the stator core **2** and disposed side by side circumferentially. Each first slot **20** extends radially outward from an inner circumference part **2s** of the stator core **2** to open radially inward.

[0053] As shown in FIG. **3**, the stator core **2** further has a plurality of in-core flow paths **22** each formed radially outside a corresponding one of the first slots **20** and axially penetrating the stator core **2**. Each in-core flow path **22** has a substantially rectangular cross-sectional surface orthogonal to the axial direction and has a circumferential width set to correspond to a circumferential width of the corresponding one of the first slots **20**.

[0054] The in-core flow paths **22** include in-core flow paths **22F** through which the oil flows axially forward and in-core flow paths **22R** through which the oil flows axially rearward. The in-core flow paths **22F** and **22R** are alternately arranged, two by two, circumferentially. For the sake of description, matters common to the in-core flow paths **22F** and **22R** may be described hereinafter by using common reference numeral **22**. The flow patterns of the oil passing through the in-core flow path **22** will be described later together with flow patterns of the oil passing through the oil guide **3A**.

[0055] As shown in FIGS. **2** and **3**, the oil guide **3A** includes an oil guide **3F** and an oil guide **3R** arranged on both sides in the axial direction of the stator core **2**. The oil guide **3F** is arranged on one side, i.e., on a front side, in the axial direction of the stator core **2**. The oil guide **3R** is arranged on the axially other end side, i.e., on a rear side, of the stator core **2**. The oil guides **3F** and **3R** each have a ring shape having substantially the same outer diameter and inner diameter as those of the stator core **2** and are each arranged coaxially with the stator core **2**. Both the oil guides **3F** and **3R** communicate via the in-core flow paths **22** with each other such that the oil passes through the in-core flow paths **22** of the stator core **2** from one of the oil guides to the other of the oil guides.

[0056] The oil guides **3F** and **3R** are arranged to face away from each other and have the same shape. Therefore, for the sake of description, matters common to the oil guides **3F** and **3R** may be described hereinafter by using common reference character **3A**.

[0057] The oil guide **3A** is made of non-magnetic material, more specifically, resin material (e.g., syndiotactic polystyrene (SPS), polyphenylene sulfide (PPS)). As shown in FIG. **3**, the oil guide **3A** includes a plurality of second teeth **31** formed over the entire axial length of the oil guide **3A**. The

plurality of second teeth **31** extend circumferentially inward to correspond one by one to the plurality of first teeth **21** of the stator core **2** and are aligned side by side circumferentially. A space between adjacent ones of the plurality of second teeth **31** is a second slot **30**. Thus, the oil guide **3A** has a plurality of second slots **30** axially penetrating the oil guide **3A** to correspond one by one to the plurality of first slots **20**. Each second slot **30** extends radially outward from an inner circumference part **3s** of the oil guide **3A** to open radially inward.

[0058] Referring back to FIG. 2, the coils **4** are so-called segment conductor (SC) windings including a plurality of segment coils connected to each other. Although not shown, the coils **4** are included in a plurality of (in the present embodiment, three) coil groups (corresponding to three phases) through which currents with different phases flow, and each coil **4** has an outer surface covered with an insulation coating.

[0059] FIG. 4 is an enlarged view of the coil ends **40**. As shown in FIG. 2, the coils **4** are attached to the stator core **2** and the oil guide **3A** such that the plurality of coil ends **40** axially protrude from the oil guide **3A**. More specifically, the coils **4** are inserted into the plurality of first slots **20** of the stator core **2** and the plurality of second slots **30** of the oil guide **3A**. Thus, the coil ends **40** axially protrude from the second slots **30** as shown in FIG. 4. The coils **4** are thus attached to the stator core **2** and the oil guide **3A**, and thereby, the oil guide **3A** is arranged in the proximity of the coil ends **40** of the coils **4**.

[0060] The arrangement and the shape of each coil end **40** depend on the orientation (extension direction) of a corresponding one of the coils **4**. As an example, some (coil ends **40a**, **40b**, and **40c**) of the coil ends **40** shown in FIG. 4 will be described. Note that in FIG. 4, the coil ends **40a**, **40b**, and **40c** arranged on an upper side are shown, and these coil ends **40a**, **40b**, and **40c** are repeatedly arranged over the entire circumference.

[0061] As shown in FIG. 4, the coil ends **40a**, **40b**, and **40c** are arranged to overlap each other in this order radially from outside to inside. Each of the coil end **40b** and the coil end **40c** overlapped radially inside by the coil end **40b** has a tilted part **41** extending obliquely outward in a direction intersecting the axial direction in accordance with the orientation of a corresponding one of the coils **4**. An inside part **42** in the axial direction of the tilted part **41** faces the oil guide **3A**. Further, the coil end **40a** has a step part **43** which rises radially outward by one step so as not to interfere with the coil end **40b** which is radially inward of and overlapping the coil end **40a**. Thus, the coil ends **40** are variously arranged and have various shapes so as not to interfere with each other while the coil ends **40** are in accordance with the orientations of the respective coils **4**.

[0062] Referring back to FIGS. 2 and 3, the sub-oil guide **5** includes a sub-oil guide **5F** and a sub-oil guide **5R** which are arranged axially outside the oil guide **3A**. The sub-oil guide **5F** is arranged forward of the oil guide **3F**. The sub-oil guide **5R** is arranged rearward of the oil guide **3R**. The sub-oil guides **5F** and **5R** each have a cylindrical shape and are arranged coaxially with the oil guides **3F** and **3R**.

[0063] The sub-oil guides **5F** and **5R** are arranged to face away from each other and have the same shape. Therefore, for the sake of description, matters common to the sub-oil guides **5F** and **5R** may be described hereinafter by using common reference numeral **5**.

[0064] The sub-oil guide **5** is made of non-magnetic material, more specifically, resin material (e.g., syndiotactic polystyrene (SPS), polyphenylene sulfide (PPS)). The sub-oil guide **5** has an outer circumference part **5p** having a guide groove **50** and a cutout **51**. The guide groove **50** is configured to receive the externally supplied oil and guide the oil to the oil guide **3A**. The cutout **51** is configured to discharge the oil, which has been used to cool the coil ends **40**, from the stator **12**.

[0065] As shown in FIG. 3, the guide groove **50** includes a circularly annular groove **500** and a straight groove **501**. The circularly annular groove **500** has an axially inner end which lowers radially inward by one step toward the oil guide **3A**. The straight groove **501** extends axially outward from an upper part of the annular groove **500**. The oil flowing from the flow path **90** of the housing **9** in FIG. 1 is dropped onto the straight groove **501** and then flows from the circularly

annular groove **500** to the oil guide **3A** (specifically, an introduction part **33A** which will be described in detail later), and the oil is used to cool the coil **4** and is then discharged from an inner circumference side of the sub-oil guide **5** through the cutout **51**.

Configuration of Oil Guide

[0066] FIG. **5** is a schematic front view of part of the oil guide **3A** of the first embodiment. FIG. **6** is a schematic perspective view of the part of the oil guide **3A** of FIG. **5** viewed from an axially inner side. Then, the detailed configuration of the oil guide **3A** will be described together with the flow patterns of the oil passing through the oil guide **3A**.

[0067] As shown in FIGS. **5** and **6**, the oil guide **3A** includes, in addition to a chamber **32A** as an interior space and the introduction part **33A** to which the externally supplied oil is introduced, a plurality of ejection parts **34**, a plurality of first connectors **35** (core connectors), a plurality of second connectors **36**, and a plurality of sub-ejection parts **37**. Each ejection part **34** communicates via the chamber **32A** with the introduction part **33A** and is configured to eject the oil toward the coil end **40**. Each first connector **35** communicates via the chamber **32A** with the introduction part **33A** and is connected to an inflow port of the in-core flow path **22**, and each second connector **36** is connected to an outlet port of the in-core flow path **22** in the stator core **2**. Each sub-ejection part **37** communicates with a corresponding one of the second connectors **36** and is configured to eject the oil flowing from another oil guide **3A** toward the coil end **40**. The flow patterns of the oil passing through the oil guide **3A** including these components are classified into the following three patterns.

Flow Pattern of Oil

[0068] FIG. **7** is a view illustrating a first flow pattern. Note that in FIG. **7**, the coils **4** are not shown for the sake of illustration. FIG. **8** is a view illustrating a flow of oil in the oil guide **3A** in the first flow pattern. As shown in FIG. **7**, the first flow pattern is a pattern of the flow of the oil through the same oil guide **3A**, and as shown in FIG. **8**, the externally supplied oil is introduced to the introduction part **33A** of the oil guide **3A** and then passes through the chamber **32A**, and the ejection parts **34** of the same oil guide **3A** eject the oil toward the coil ends **40** in the proximity thereof. Note that FIGS. **7** and **8** show the flow of the oil in the front oil guide **3F**, and the flow of the oil in the rear oil guide **3R** is the same as that in the front oil guide **3F** except that the direction of the oil ejected from the ejection parts **34** is opposite to the direction (facing the viewer) shown in FIG. **7**.

[0069] In the first flow pattern as described above, the externally supplied oil introduced to the introduction part **33A** by the introduction part **33A** and the ejection parts **34** of the oil guide **3A** is then ejected as it is from the ejection parts **34** toward the coil ends **40** in the proximity of the ejection parts **34**. Therefore, the externally supplied oil is ejected onto the coil ends **40** in a fresh state, a state where the externally supplied oil has not drawn heat from another component (e.g., the stator core **2**) of the stator **12**. Thus, the coils **4** can be efficiently cooled.

[0070] FIG. **9** is a view illustrating a second flow pattern. Note that in FIG. **9**, the coils **4** are not shown for the sake of illustration, and the first connectors **35** and the second connectors **36** do not appear. FIG. **10** is a view illustrating a flow of oil in the front oil guide **3F** in the second flow pattern. FIG. **11** is a view illustrating a flow of oil in the rear oil guide **3R** in the second flow pattern. As shown in FIG. **9**, the second flow pattern is a pattern of oil flowing from the front oil guide **3F** to the rear oil guide **3R**. More specifically, in the second flow pattern, the externally supplied oil is introduced to the introduction part **33A** of the front oil guide **3F** as shown in FIGS. **9** and **10**, then passes through the chamber **32A** and the first connectors **35** of the oil guide **3F** as shown in FIG. **10**, and passes the in-core flow paths **22R** of the stator core **2** as shown in FIG. **9**. Then, the externally supplied oil further passes through the second connectors **36** in the rear oil guide **3R** as shown in FIGS. **9** and **11** and is ejected from the sub-ejection parts **37** of the oil guide **3R** toward the coil ends **40** on the rear side.

[0071] In the second flow pattern as described above, the first connectors **35** of the front oil guide

3F allow part of the externally supplied oil introduced to the introduction part 33A of the oil guide 3F to flow from the first connectors 35 into the in-core flow paths 22R of the stator core 2. Thus, the stator core 2 can be cooled.

[0072] Further, in the second flow pattern, the sub-ejection parts 37 of the rear oil guide 3R eject the oil passing from the oil guide 3F through the in-core flow paths 22R of the stator core 2 toward the coil ends 40 on the rear side. This enables the oil supplied for cooling the stator core 2 to be used without waste to cool the coil ends 40 on the rear side.

[0073] FIG. 12 is a view illustrating a third flow pattern. Note that in FIG. 12, the coils 4 are not shown for the sake of illustration, and the first connectors 35 and the second connectors 36 do not appear. FIG. 13 is a view illustrating a flow of oil in the front oil guide 3F in the third flow pattern. FIG. 14 is a view illustrating a flow of oil in the rear oil guide 3R in the third flow pattern. As shown in FIG. 12, the third flow pattern is a pattern of oil flowing from the rear oil guide 3R to the front oil guide 3F in contrast to the second flow pattern described above. More specifically, in the third flow pattern, the externally supplied oil is introduced to the introduction part 33A of the rear oil guide 3R as shown in FIGS. 12 and 14, then passes through the chamber 32A and the first connectors 35 of the oil guide 3R as shown in FIG. 14, passes through the in-core flow paths 22F of the stator core 2 as shown in FIG. 12, further passes through the second connectors 36 of the front oil guide 3F as shown in FIGS. 12 and 13, and is ejected from the sub-ejection parts 37 of the oil guide 3F toward the coil ends 40 on the front side.

[0074] Similarly to the second flow pattern, such a third flow pattern allows part of the externally supplied oil to flow from the first connectors 35 of the oil guide 3R into the in-core flow paths 22F of the stator core 2, thereby cooling the stator core 2. Further, the oil passing through the in-core flow paths 22F is ejected from the sub-ejection parts 37 of the oil guide 3F toward the coil ends 40 on the front side, and thereby, the oil supplied for cooling the stator core 2 is used without waste to cool the coil ends 40 on the front side.

[0075] Components of the oil guide 3A implementing the flow patterns are configured as described below in detail.

Details of Components of Oil Guide

[0076] Referring back to FIG. 5, the chamber 32A includes an annular part 320 and a plurality of branches 321. The annular part 320 is arranged to surround the second slots 30 from radially outside and is concentric. The plurality of branches 321 branch radially inward from the annular part 320 toward the interior of the second teeth 31. Each branch 321 is connected to a corresponding one of the ejection parts 34, and therefore, the chamber 32A is connected to the ejection parts 34 at the respective branches 321. In the annular part 320, a plurality of spacers 322 having an axial thickness are circumferentially provided with a space therebetween. This can suppress the chamber 32A from being axially compressed, that is, the oil guide 3A from axially deforming.

[0077] Further, as described above, the ejection parts 34 communicate via the chamber 32A including the annular part 320 with the introduction part 33A, and thereby, the oil in the chamber 32A can be ejected from the ejection parts 34 in the state where the annular part 320 is uniformly filled in the circumferential direction with the oil introduced to the introduction part 33A. This enables the oil to be circumferentially uniformly ejected from the ejection parts 34, and thus, the oil can be circumferentially uniformly applied to each of the coil ends 40.

[0078] As shown in FIG. 3, the introduction part 33A is open radially outward over the entire circumference of an outer circumference part 3p of the oil guide 3A. Thus, the oil can be introduced into the oil guide 3A from the entire circumference of the outer circumference part 3p of the oil guide 3A.

[0079] As shown in FIGS. 4 and 5, the ejection parts 34 are each a circular hole open axially outward and are provided in an axially outer part of the oil guide 3A. The ejection parts 34 are circumferentially arranged spaced apart from each other. As shown in FIG. 4, the ejection parts 34

are arranged in the proximity of the coil ends **40**. This enables the oil to be reliably applied to the coil ends **40**.

[0080] Further, as shown in FIG. 5, the ejection parts **34** are arranged on the second teeth **31**, that is, are each arranged between adjacent ones of the second slots **30** and include ejection parts **34a** arranged radially outside and ejection parts **34b** arranged radially inside. The ejection parts **34a** and the ejection parts **34b** are arranged on the respective second teeth **31** such that the ejection parts **34a** and the ejection parts **34b** are circumferentially aligned two by two alternately. The ejection parts **34** are thus each arranged between adjacent ones of the second slots **30**, and thereby, regardless of the radial position of the coil ends **40** of the coils **4** inserted into the first slots **20** and the second slots **30**, for example, the ejection parts **34** can be arranged in the proximity of the coil ends **40** such that the ejection parts **34b** are arranged in the proximity of the coil ends **40** arranged radially inside, and the oil can be ejected from the proximity of the coil ends **40**. This enables the oil to be reliably applied to the coil ends **40**.

[0081] As shown in FIGS. 5 and 6, the first connectors **35** are each a circular hole open axially inward and are provided in an axially inner part of the oil guide **3A**. The first connectors **35** of the front oil guide **3F** are connected to the inflow ports of the in-core flow paths **22R**, and likewise, the first connectors **35** of the rear oil guide **3R** are connected to the inflow ports of the in-core flow paths **22F**. Therefore, each first connector **35** is arranged at a circumferential position and a radial position corresponding to the inflow port of an associated one of the in-core flow paths **22**. The opening area of each first connector **35** is set to be smaller than the opening area of the inflow port of the associated one of the in-core flow paths **22**. Thus, the inflow pressure of the oil to the in-core flow paths **22** can be increased.

[0082] Also shown in FIGS. 5 and 6, the second connectors **36** are each an elongated hole which is open axially inward and which has a circumferential length corresponding to the circumferential width of the outlet port of a corresponding one of the in-core flow paths **22**, and each second connector **36** is provided in the axially inner part of the oil guide **3A**. The second connectors **36** of the front oil guide **3F** are connected to the outflow ports of the in-core flow paths **22F**, and likewise, the second connectors **36** in the rear oil guide **3R** are connected to the outflow ports of the in-core flow paths **22R**. Thus, each second connector **36** is arranged at a circumferential position and a radial position corresponding to the outflow port of an associated one of the in-core flow paths **22**. The opening area of each second connector **36** is set to be substantially equal to the opening area of the outlet port of the associated one of the in-core flow paths **22**. As shown in FIG. 6, the first connectors **35** and the second connectors **36** are circumferentially arranged two by two alternately corresponding to the alignment of the in-core flow paths **22F** and **22R**.

[0083] As shown in FIGS. 4 and 5, the sub-ejection parts **37** are each a circular hole open axially outward and are provided in an axially outer part of the oil guide **3A**. The sub-ejection parts **37** are circumferentially arranged spaced apart from each other at circumferential positions corresponding to the second connectors **36** and radially outside the second slots **30**. Each sub-ejection part **37** has a smaller opening area than a corresponding one of the second connectors **36** and is connected via a flow path **370** to the corresponding one of the second connectors **36**. The flow path **370** has a cross-sectional area which is orthogonal to its extension direction (in the present embodiment, the axial direction) and which gradually decreases from the second connector **36** toward the sub-ejection part **37**. This enables the outflow pressure of the oil from the sub-ejection part **37** to be increased.

[0084] Note that similarly to the flow path **370** connected to the sub-ejection part **37**, flow paths each connected to a corresponding one of the ejection parts **34** (flow paths each leading to a corresponding one of the ejection parts **34** and the chamber **32A**) may have a cross-sectional area which is orthogonal to its extension direction and which gradually decreases from the chamber **32A** toward the corresponding one of the ejection parts **34**. Similarly to the above-described case, this enables the outflow pressure of the oil from the ejection part **34** to be increased.

[0085] In addition to the above-described configuration, as shown in FIGS. 5 and 6, the oil guide

3A has a plurality of insertion claws **38** which are to be inserted into the inflow ports of the in-core flow paths **22** to position the oil guide **3A** with respect to the stator core **2**. The insertion claws **38** are each a projection protruding axially inward from the axially inner part of the oil guide **3A** and are arranged on circumferentially both sides of one first connector **35** of two successively aligned first connectors **35**. Thus, the oil guide **3A** is provided with the insertion claws **38**, and therefore, the oil guide **3A** can be positioned with respect to the stator core **2** with a simple configuration. [0086] The ejection parts **34** and the sub-ejection parts **37** are each configured to eject the oil toward the coil end **40**, wherein these parts are configured such that the oil is applied to individual coil ends **40**. More specifically, the ejection parts **34** and the sub-ejection parts **37** are configured such that different coil ends **40** are their respective ejection targets, that is, the ejection parts **34** and the sub-ejection parts **37** are configured to eject the oil toward respective ones of the coil ends **40**. Thus, the oil is applied to each of the coil ends **40**, and thus, the entirety of the coils **4** can be more efficiently cooled.

[0087] Further, the ejection parts **34** and the sub-ejection parts **37** are configured as described in further detail below so that the oil is ejected appropriately to the arrangement and the shape of each of the coil ends **40** which are the ejection targets.

Ejection Manner of Ejection Part and Sub-Ejection Part

[0088] FIG. **15** is a view for explaining an ejection manner of the ejection parts **34** and the sub-ejection parts **37**. FIG. **16** is an enlarged view of the ejection parts **34a** on a radially outside and the coil ends **40b** which are ejection targets of the ejection parts **34a**. Note that in FIG. **16**, the coil ends **40a** radially outward of and overlapping the coil ends **40b** is omitted for the sake of illustration. FIG. **17** is an enlarged view of the ejection parts **34b** on radially inside and the coil ends **40c** which are ejection targets of the ejection parts **34b**. Note that for the sake of illustration, the coil ends **40b** radially outward of and overlapping the coil ends **40c** and the coil ends **40a** radially outward of and overlapping the coil ends **40b** are omitted in FIG. **17**.

[0089] As an example of the ejection manner, the ejection targets of the ejection parts **34a** (ejection parts on the radially outside) indicated by being surrounded by an ellipse P1 of FIG. **15** are the coil ends **40b** shown in FIG. **16**, and the ejection parts **34a** are configured to eject the oil toward the inside parts **42** facing the oil guide **3A** at the tilted parts **41** of the coil ends **40b**. Similarly, the ejection targets of the ejection parts **34b** (ejection parts on the radially inside) indicated by being surrounded by an ellipse P2 of FIG. **15** are the coil ends **40c**, shown in FIG. **17**, radially inward of and overlapping the coil ends **40b**, and the ejection parts **34b** are configured to eject the oil toward the inside parts **42**, facing the oil guide **3A**, of the tilted parts **41** of the coil ends **40c**.

[0090] The ejection parts **34a** and **34b** having the above-described configuration eject the oil toward the inside parts **42** of the tilted parts **41** of the coil ends **40b** and **40c**, that is, parts in the proximity of the oil guide **3A**. Thus, the oil can be reliably applied to the coil ends **40b** and **40c**. Further, as an accompanying effect, when the oil is applied to the tilted parts **41** of the coil ends **40b** and **40c**, the oil moves axially outward along the tilted parts **41**, thereby more efficiently cooling the coil ends **40b** and **40c**.

[0091] Further, as shown in FIGS. **16** and **17**, the ejection parts **34a** and **34b** are, in particular, configured to eject the oil radially outward of the coil ends **40b** and **40c**. Thus, at the coil ends **40b** and **40c** arranged on a side corresponding to the 12 o'clock side (an upper side) of the oil guide **3A**, that is, the coil ends **40b** and **40c** arranged on the upper side of the oil guide **3A**, the oil moves downward along the coil ends **40b** and **40c** due to its self-weight, thereby more efficiently cooling the coil ends **40b** and **40c**.

[0092] FIG. **18** is an enlarged view of the sub-ejection parts **37** and the coil ends **40a** which are ejection targets of the sub-ejection parts **37**. FIG. **19** is a schematic longitudinal sectional view of the coil end **40a** arranged on a lower side of the oil guide **3A** and peripheral parts of the coil end **40a**. In addition to the examples described above, the ejection targets of the sub-ejection parts **37** indicated by being surrounded by an ellipse P3 of FIG. **15** are the coil ends **40a** shown in FIG. **18**,

and the sub-ejection parts **37** are configured to eject the oil radially inward from radially outside the coil ends **40**. More specifically, as shown in FIG. **19**, each flow path **370** leading to a corresponding one of the sub-ejection parts **37** is tilted radially inward (in FIG. **19**, upward) toward the corresponding one of the sub-ejection parts **37**.

[0093] The sub-ejection parts **37** having the above-described configuration eject the oil radially inward toward the step parts **43** of the coil ends **40a** present radially inside the sub-ejection parts **37**. In particular, as shown in FIG. **19**, toward the coil ends **40a** arranged on a side corresponding to the 6 o'clock side (a lower side) of the oil guide **3A**, that is, the stepped parts **43** of the coil ends **40a** located upward of the sub-ejection parts **37**, the oil is ejected upward against the gravity. Thus, the oil can reliably be applied to the coil ends **40a** located radially inside the sub-ejection parts **37**. [0094] Note that although illustration is omitted, some of the ejection parts **34** may be configured to eject the oil radially inward from radially outside the coil ends **40** in a similar manner to the sub-ejection parts **37**, and also in this case, the oil can be reliably applied to the coil ends **40** located radially inside the ejection parts **34** in a similar manner to the above-explained case.

Configuration of Housing

[0095] Referring back to FIG. **1**, the housing **9** includes: a body part **91** having a circularly cylindrical shape; and a pair of lids **92** each of which has a bottomed circularly cylindrical shape and which are arranged to close respective openings on axial ends of the body part **91**. The bottom of each lid **92** has a hole **920** into which the shaft **10** is to be inserted. Each lid **92** is fixed to a corresponding one of the axial ends of the body part **91** with a fastener (not shown) such as a bolt.

[0096] In a state where each of the lids **92** is fixed to the body part **91**, a bottom inner surface **92a** of the each of the lids **92** is in contact with the sub-oil guide **5** arranged at the axial end of the stator **12** and presses the sub-oil guide **5** axially inward. In this state, the oil guide **3A** arranged axially inside the sub-oil guide **5** is pressed by the lid **92** axially against the stator core **2** via the sub-oil guide **5**. That is, the housing **9** is configured to press the oil guide **3A** of the stator **12** axially against the stator core **2**. With the housing **9** having the above-described configuration, the oil is suppressed from leaking from a location between the stator core **2** and the oil guide **3A** without providing a sealing member, such as an O-ring, between the stator core **2** and the oil guide **3A**. This can reduce the number of components included in the motor unit A, thereby reducing the cost of the motor unit A.

[0097] With the stator **12** described above, the oil guide **3A** provided with the introduction part **33A** and the ejection parts **34** provides the effect that the externally supplied oil is ejected onto the coil ends **40** in a fresh state, a state where the externally supplied oil has not drawn heat from another component of the stator **12**, and thus, the coils **4** can be efficiently cooled.

[0098] Further, an accompanying effect in addition to the above effect is that configuring the oil guide **3A** as a component separate from the stator core **2** enables the oil guide **3A** to be provided for the stator **12** without influencing the strength and/or the manufacturability of the stator core **2**.

[0099] Further, as described above, the oil guide **3A** is made of resin material, and thereby, the degree of freedom of processing the oil guide **3A** can be increased, and therefore, the arrangement and the shape of the ejection parts **34** can be freely set such that the oil is ejected in accordance with the attachment posture of the coils **4** to the stator core **2** and the oil guide **3A**, more specifically, in a manner appropriate to the arrangement and/or the shape of the coil ends **40** being the ejection targets.

[0100] Moreover, as described above, the sub-oil guide **5** has the guide groove **50**, and therefore, the externally supplied oil is guided along the guide groove **50** formed in the outer circumference part **5p** of the sub-oil guide **5** to the introduction part **33A** of the oil guide **3A**. Thus, without increasing the size of the stator **12** radially outward beyond the oil guide **3A**, the externally supplied oil can be guided to the introduction part **33A**, and thus, the oil can be introduced to the introduction part **33A** with a simple and space-saving structure.

[0101] Note that liquid other than the oil may be employed as the cooling medium. Moreover, the

oil guide **3A** may be provided at least at an axially one end of the stator core **2**. Also in this case, the effect is that the coils **4** can be efficiently cooled in a similar manner to the above-explained case. Moreover, the oil guides **3F** and **3R** may have shapes different from each other.

[0102] Note that the oil guide **3A** may be made of non-magnetic material, such as ceramics, other than resin material. Making the oil guide **3A** of non-magnetic material allows the oil guide **3A** to be provided for the stator **12** without magnetically influencing the motor **1**.

[0103] Note that one sub-oil guide **5** may be provided on one side in the axial direction in accordance with the shape of the oil guide **3A** and/or the entire layout of the motor **1** including the stator **12**, or no sub-oil guide **5** may be provided.

[0104] Note that the arrangement and/or the shape of the ejection parts **34** and the sub-ejection parts **37** are not limited to the above-described examples but may be accordingly set such that the oil is ejected in accordance with the attachment posture of the coils **4** to the stator core **2** and the oil guide **3A**, more specifically, in a manner appropriate to the arrangement and/or the shape of the coil ends **40** being the ejection targets. For example, some of the ejection parts **34** may be arranged radially outside the coil ends **40**, and some of the remaining of the ejection parts **34** may each be arranged between adjacent ones of the second slots **30**, and/or the sub-ejection parts **37** may each be arranged between adjacent ones of the second slots **30**.

[0105] Moreover, the ejection part **34** and the sub-ejection part **37** may be configured such that an identical coil end **40** is their ejection target, that is, the ejection part **34** and the sub-ejection part **37** may be configured to eject the oil toward an identical coil end **40**. Alternatively, the ejection part **34** and the sub-ejection part **37** may be configured such that two or more coil ends **40** are their ejection targets, that is, the ejection part **34** and the sub-ejection part **37** may be configured to eject the oil toward two or more coil ends **40**.

[0106] Note that it is sufficient that the sub-ejection parts **37** are configured to eject the oil flowing from an oil guide on one side toward the coil ends **40** on the other side as in the second flow pattern and the third flow pattern, and thus, the sub-ejection parts **37** may be provided at least for an oil guide on the other side (in the second flow pattern, the rear oil guide **3R**, and in the third flow pattern, the front oil guide **3F**).

[0107] Note that the attachment posture of the coils **4** to the stator core **2** and the oil guide **3A** is not limited to the example described above but may accordingly be set in accordance with the shape of the stator core **2** and the oil guide **3A**. Moreover, the coils **4** are not limited to the example described above but may be a continuous winding other than the SC windings.

Second Embodiment

[0108] FIG. **20** is a schematic front view of an oil guide **3B** of a second embodiment. Next, a difference of the configuration of the oil guide **3B** of the second embodiment of the present disclosure from the oil guide **3A** of the first embodiment will mainly be described.

[0109] As shown in FIG. **20**, the oil guide **3B** of the second embodiment may be provided with a peripheral wall **323** continuously surrounding an annular part **320** from radially outside. Here, “continuously” does not necessarily mean the entire circumference (i.e., 360 degrees) but is a concept including the case where the circumference is partially interrupted by less than about 30 degrees as shown in FIG. **20**. An upper part of the peripheral wall **323** is interrupted, and the interrupted part is an introduction part **33B** of the second embodiment. With the peripheral wall **323**, a space radially outside the annular part **320** is occupied by the peripheral wall **323**, thereby reducing the volume of a flow path in the oil guide **3B**. This can suppress air from accumulating in the oil guide **3B**, thereby suppressing the ejection pressure of the oil from the oil guide **3B** from decreasing.

[0110] FIG. **21** is an enlarged view of an upper part of the oil guide **3B** of FIG. **20**. As shown in FIG. **21**, branches **324** of the second embodiment are each connected to an ejection part **34** or a first connector **35**. A terminal part **325** which is a radially inside end of each branch **324** is rounded, and a terminal edge of the terminal part **325** of each branch **324** connected to the ejection part **34** is

tangent to a terminal edge of the ejection part **34**. Thus, a chamber **32B** of the second embodiment is connected to the ejection parts **34** at terminal ends of the branches **324**. The chamber **32B** connected to the ejection parts **34** in such a manner eliminates a space in each branch **324** between an end side and its part connected to a corresponding one of the ejection parts **34**. This can suppress air from accumulating at the terminal ends of the branches **324**, thereby suppressing the ejection pressure of the oil from the oil guide **3B** from decreasing.

[0111] Similarly to the case of the ejection parts **34** described above, the chamber **32B** of the second embodiment is connected to the first ejection parts **35** at terminal ends of the branches **324**. This can suppress air from accumulating at the terminal ends of the branches **324**, similarly to the case of the ejection parts **34** described above, thereby suppressing the inflow pressure of the oil to in-core flow paths **22** from decreasing.

[0112] FIG. **22** is a schematic longitudinal sectional view of one of flow paths **371** leading to respective sub-ejection parts **39** of the second embodiment. Specifically, FIG. **22** shows a cross-section along line A-A of FIG. **21**. As shown in FIG. **21**, the oil guide **3B** of the second embodiment is provided with the sub-ejection parts **39** radially outside second slots **30** in a similar manner to the oil guide **3A** of the first embodiment, wherein each sub-ejection part **39** of the second embodiment is a rectangular hole and is arranged adjacent to a corresponding one of the second slots **30**. As shown in FIG. **22**, the flow paths **371** leading to the sub-ejection parts **39** from second connectors **36** have openings **372** which are open toward the second slots **30**. With such openings **372**, coils **4** inserted through the second slots **30** face the flow paths **371** via the openings **372**, thereby increasing a contact area between the oil and the coils **4**. More specifically, part of the oil flowing through the flow paths **371** flows through the openings **372** into the second slots **30** and is applied to the coils **4** inserted into the second slots **30**. Thus, the coils **4** can be more efficiently cooled.

[0113] Note that although illustration is omitted, when the ejection parts **34** are arranged radially outside the second slots **30**, flow paths leading to the ejection parts **34** (flow paths leading to the ejection parts **34** and a chamber **32A**) may have openings in a similar manner to the flow paths **371**. Also in this case, the coils **4** can be more efficiently cooled in a similar manner to the above case.

[0114] The embodiments described above are examples in all aspects and should not be construed as limiting. Thus, the technical scope of the present invention should not be interpreted based on only the embodiments and the examples described above but is set forth based on the recitation of the claims. Moreover, modifications and variations belonging to the equivalent of the scope of the claims should all fall within the scope of the present invention.

[0115] While the present disclosure has been described with reference to embodiments thereof, it is to be understood that the disclosure is not limited to the embodiments and constructions. The present disclosure is intended to cover various modification examples and modifications within the range of equivalency. In addition, various combinations and configurations, and further, other combinations and configurations including more, less, or only a single element thereof are also within the spirit and scope of the present disclosure.

Claims

1. A stator comprising: a stator core having a circularly cylindrical shape; a guide member having a ring shape and arranged coaxially with the stator core on at least one side in an axial direction of the stator core; and coils attached to the stator core and the guide member such that a plurality of coil ends of the coils axially protrude from the guide member, wherein the guide member includes an introduction part to which a cooling medium which is externally supplied is introduced, and a plurality of ejection parts which communicate with the introduction part and which are configured to eject the cooling medium toward the plurality of coil ends.

2. The stator of claim 1, wherein some of the plurality of coil ends have tilted parts each extending

obliquely outward in a direction intersecting the axial direction, and some of the plurality of ejection parts are configured to eject the cooling medium toward inside parts of the tilted parts, the inside parts facing the guide member.

3. The stator of claim 1, wherein some of the plurality of ejection parts are configured to eject the cooling medium radially inward from radially outside the plurality of coil ends.

4. The stator of claim 1, wherein the stator core may have a plurality of first slots each of which axially penetrates the stator core and which are aligned side by side circumferentially, the guide member has a plurality of second slots each of which axially penetrates the guide member such that the plurality of second slots correspond to the plurality of first slots, the plurality of coil ends axially protrude from the plurality of second slots when the coils are inserted into the plurality of first slots and the plurality of second slots, and some of the plurality of ejection parts are arranged between adjacent ones of the plurality of second slots.

5. The stator of claim 1, wherein the plurality of ejection parts are configured to eject the cooling medium toward respective ones of the coil ends.

6. The stator of claim 1, wherein the stator core has an in-core flow path axially penetrating the stator core, and the guide member has a core connector which communicates with the introduction part and which is connected to an inflow port of the in-core flow path.

7. The stator of claim 6, wherein the guide member is a first guide member, the stator includes a second guide member arranged on an opposite side of the stator core from the first guide member in the axial direction, the first guide member and the second guide member communicate with each other via the in-core flow path such that the cooling medium passes through the in-core flow path from the first guide member to the second guide member, and the second guide member is provided with a plurality of sub-ejection parts configured to eject the cooling medium flowing from the first guide member toward coil ends which are in a proximity of the second guide member.

8. The stator of claim 7, wherein the stator core has a plurality of first slots each axially penetrating the stator core, the second guide member has a plurality of second slots corresponding, one by one, to the plurality of first slots, the coils are inserted into the plurality of first slots and the plurality of second slots of the second guide member, each of the plurality of sub-ejection parts is arranged radially outside the second slots of the second guide member, and flow paths connected to the plurality of sub-ejection parts have openings which are open toward the plurality of second slots of the second guide member.

9. The stator of claim 1, wherein the guide member has a chamber including an annular part which is concentric, and the plurality of ejection parts communicate with the introduction part via the chamber.

10. The stator of claim 9, wherein the guide member is provided with a peripheral wall continuously surrounding the annular part from radially outside.

11. The stator of claim 9, wherein the chamber includes a plurality of branches branched radially inward from the annular part, and the branches have ends connected to the plurality of ejection parts.

12. The stator of claim 1, wherein the guide member is made of non-magnetic material.

13. The stator of claim 1, wherein the guide member is made of resin material.

14. The stator of claim 1, further comprising a sub-guide member arranged axially outside the guide member, being coaxial with the guide member, and having a cylindrical shape, wherein the introduction part of the guide member is open radially outward, and the sub-guide member has an outer circumference part having a guide groove configured to: receive the cooling medium which is externally supplied; and guide the cooling medium to the introduction part.

15. A motor unit comprising: a motor including the stator of claim 1; and a housing in which the stator is housed, wherein the housing is configured to press the guide member of the stator axially against the stator core.
